



SUMMARY OF PRODUCT CHARACTERISTICS

BETAQUIL
Betahistine Dihydrochloride
Tablets BP 8 mg and 16 mg
R_x Only

NAME OF DRUG PRODUCT: Betahistine Dihydrochloride Tablets BP 8 mg
Betahistine Dihydrochloride Tablets BP 16 mg

(TRADE) NAME OF PRODUCT: BETAQUIL 8
BETAQUIL 16
STRENGTH: 8 mg and 16 mg

PHARMACEUTICAL DOSAGE FORM: Tablet.

QUALITATIVE AND QUANTITATIVE COMPOSITIONS:

Betahistine Dihydrochloride Tablets BP 8 mg:
Each un-coated tablet contains Betahistine Dihydrochloride Ph.Eur. 8 mg.
Betahistine Dihydrochloride Tablets BP 16 mg:
Each un-coated tablet contains Betahistine Dihydrochloride Ph.Eur. 16 mg.

PHARMACEUTICAL FORM:

Betahistine Dihydrochloride Tablets BP 8 mg:
White to off-white round, flat uncoated tablets debossed with 'X' on one side and '87' on the other side.
Betahistine Dihydrochloride Tablets BP 16 mg:
White to off-white round uncoated tablets debossed with 'X' and a break line on one side and '88' on the other side.

CLINICAL PARTICULARS:

Therapeutic indications

Meniere's Syndrome as defined by the following core symptoms:

- Vertigo (with nausea/vomiting)
- Hearing loss (hardness of hearing)
- Tinnitus (ringing in the ears)
- Symptomatic treatment of vestibular vertigo

Posology and method of administration

Always take Betahistine dihydrochloride tablets exactly as your doctor has prescribed. If you have any questions, contact your doctor or pharmacist. If you forget to take your tablet(s), do not take a double dose to compensate for it. If you require further information, please ask your doctor or pharmacist for advice. The dosage for adults is 8 mg – 16 mg three times daily. Your doctor will adjust the dosage according to your response to the medication. Improvement of symptoms may take up to two weeks and the best results are sometimes obtained only after a few months. There are indications that treatment from the onset of the disease prevents its progression and/or the loss of hearing in later phases of the disease. Paediatric population: Betahistine dihydrochloride tablets is not recommended for use in children under the age of 18 years due to insufficient data on safety and efficacy. Geriatric population: No dose adjustment is required in elderly. Renal impairment: No dose adjustment is required in renal impaired patients. Hepatic impairment: No dose adjustment is required in hepatic impaired patients.

Contraindications

Do not take Betahistine Dihydrochloride Tablets if you are hypersensitive (allergic) to the active substance or to any of the excipients.

Special warnings and precautions for use

If you suffer a phaeochromocytoma (an adrenal gland tumor) or bronchial asthma, your doctor will need to monitor you carefully while you are taking this medication. Furthermore, please inform your doctor or pharmacist if you have a history of peptic (stomach) ulcer before taking this medication.

Interaction with other medicinal products and other forms of interaction

Please tell your doctor or pharmacist if you are taking or have recently taken any other medicines including medicines obtained without a prescription. No in vivo interaction studies have been performed. Based on in vitro data, no in vivo on Cytochrome P450 enzymes is expected. In vitro data indicate an inhibition of betahistine metabolism by drugs that inhibit monoamine-oxidase (MAO) including MAO subtype B (e.g. selegiline). Caution is recommended when using betahistine and MAO inhibitors (including MAO-B selective) concomitantly.

Pregnancy and lactation

Ask your doctor or pharmacist for advice before taking any medicine during pregnancy.

Pregnancy:

There is insufficient data on the use of betahistine in pregnant women. Animal studies are insufficient with respect to effects on pregnancy, embryonal/fetal development, parturition (giving birth) and postnatal development. The potential risk for humans in this regard is unknown. Betahistine should not be used during pregnant unless is it deemed absolutely necessary by your doctor.

Lactation:

It is not known whether betahistine is excreted in human milk. There are no animal studies on the excretion of betahistine in milk. You should not take Betahistine if you are nursing. For further information, talk to your doctor regarding the importance of this medicine to you, the benefits of nursing and the potential risks to your child.

Effects on ability to drive and use machines

Betahistine is not likely to affect your ability to drive or use tools or machinery. However, remember that the disease for which you are being treated with Betahistine (Ménière's disease) can make you feel dizzy or sick, and can affect your ability to drive and use machines.

Undesirable effects

Like all medicines, Betahistine dihydrochloride tablets may have side effects. If you notice any side effects not mentioned in this leaflet, or if any of the side effects get serious, please inform your doctor or pharmacist.

The following undesirable effects have been experienced with the below indicated frequencies in betahistine-treated patients in placebo-controlled clinical trials: very common (≥1/100 to <1/10); common (≥1/100 to <1/10); uncommon (≥1/1000 to <1/100); rare (≥1/10 000 to <1/1000); very rare (<1/10 000).

Gastrointestinal disorders

Common: nausea and dyspepsia (indigestion)
In addition to those events reported during clinical trials, the following undesirable effects have been reported spontaneously during postmarketing use and in scientific literature. A frequency cannot be estimated from the available data and is therefore classified as 'not known'.

Immune system disorders

Hypersensitivity (allergic) reactions (such as anaphylaxis) have been reported.

Gastrointestinal disorders

Mild gastric complaints (e.g. vomiting, gastrointestinal pain, abdominal distension and bloating) have been observed. These can normally be dealt with by taking the dose during meals or by lowering the dose.

Skin and subcutaneous tissue disorders

Cutaneous (skin) and sub cutaneous (tissue under the skin), hypersensitivity (allergic) reactions have been reported, in particular angioneurotic oedema (sudden onset of face, neck or limb swelling), urticaria (hives), rash and pruritus (itchiness).

Overdosage

Symptoms of overdose:

A few overdose cases have been reported. Some patients experienced mild to moderate symptoms such as nausea, somnolence (sleepiness) and abdominal pain with doses up to 640 mg. More serious complications including convulsions, and pulmonary (lung) and cardiac (heart) complications were observed in cases of intentional overdose of Betahistine dihydrochloride tablets, especially when taken in combination with other overdosed drugs.

Treatment of overdose:

No specific antidote is known. Treatment of overdose should include standard supportive measures.

P-1532759

■ Black A/s: 165 x 210 mm

 AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		BETAQUIL	Leaflet	P1532759	04.01.2023 & 12.00 PM
		Customer / Country	Version No.	Reason Of Issue	Reviewed / Approved by
		Malaysia_U3	01	Revision	
Team Leader	Kiran.k	Dimensions	No. of Colours : 01		
Initiator	Shirisha	165 x 210 mm	 32759		
Artist:	SCD	Pharmacode			
		32759			
Additional Information : Supersede Code : P1513163					Sign / Date

	PHARMACOLOGICAL PROPERTIES
	Pharmacodynamic properties
	Pharmacotherapeutic group: Anti-vertigo preparations.
	The mechanism of action of betahistine is only partly understood. There are several plausible hypotheses that are supported by animal studies and human data:
	<u>Betahistine affects the histaminergic system</u>
	Betahistine acts both as a partial histamine H1-receptor agonist and histamine H3-receptor antagonist also in neuronal tissue, and has negligible H2-receptor activity.

Betahistine increases histamine turnover and release by blocking presynaptic H3-receptors and inducing H3-receptor downregulation.

Betahistine may increase blood flow to the cochlear region as well as to the whole brain:

Pharmacological testing in animals has shown that blood circulation in the striae vascularis of the inner ear improves, probably by means of a relaxation of the precapillary sphincters of the microcirculation of the inner ear.

Betahistine was also shown to increase cerebral blood flow in humans.

Betahistine facilitates vestibular compensation:

Betahistine accelerates the vestibular recovery after unilateral neurectomy in animals, by promoting and facilitating central vestibular compensation; this effect, characterized by an up-regulation of histamine turnover and release, is mediated through H3 Receptor antagonism, in human subjects, recovery after vestibular neurectomy was also reduced when treated with betahistine.

Betahistine alters neuronal firing in the vestibular nuclei:

Betahistine was also found to have a dose dependent inhibiting effect on spike generation of neurons in lateral and medial vestibular nuclei. The pharmacodynamics properties as demonstrated in animals may contribute to the therapeutic benefit of betahistine in the vestibular system.

The efficacy of betahistine was shown in studies in patients with vestibular vertigo and with Meniere's disease as was demonstrated by improvements in severity and frequency of vertigo attacks.

Pharmacokinetic properties

Absorption

Orally administered betahistine is readily and almost completely absorbed from all parts of the gastro-intestinal tract. After absorption, the drug is rapidly and almost completely metabolized into 2-pyridylacetic acid. Plasma levels of betahistine are very low. Pharmacokinetics analyses are therefore based on 2-PAA measurements in plasma and urine.

Under fed conditions C_{max} is lower compared to fasted conditions. However, total absorption of betahistine is similar under both conditions, indicating that food intake only slows down the absorption of betahistine.

Distribution

The percentage of betahistine that is bound by blood plasma proteins is less than 5%.

Biotransformation

After absorption, betahistine is rapidly and almost completely metabolized into 2-PAA (which has no pharmacological activity). After oral administration of betahistine the plasma (and urinary) concentration of 2-PAA reaches its maximum 1 hour after intake and declines with a half life of about 3.5 hours.

Excretion

2-PAA is readily excreted in the urine. In the dose range of 7 to 48 mg, about 85% of the original dose is excreted in the urine. Renal or fecal excretion of betahistine itself is of minor importance.

Linearity

Recovery rates are constant over the oral dose range of 8-48 indicating that the pharmacokinetics of betahistine are linear, and suggesting that the involved metabolic pathway is not saturated.

PHARMACEUTICAL PARTICULARS

List of excipients

Cellulose microcrystalline (Avicel PH 101), Mannitol, Povidone, Croscopovidone, Citric acid, anhydrous, Cellulose microcrystalline (Avicel PH 112), Silica colloidal anhydrous, Talc and Stearic acid.

Incompatibilities

Not applicable.

Shelf life

Please refer outer package for expiry date.

Special precautions for storage

Do not store above 30 °C.

Protect from light and moisture.

Nature and contents of container

10x10's Blister Pack

Manufactured by:



AUROBINDO
AUROBINDO PHARMA LIMITED-UNIT III

Survey no.: 313, 314, Bachupally,
Bachupally Mandal, Medchal-Malkajgiri District,
Telangana State, 500090, India

Product Registration Holder

Healol Pharmaceuticals Sdn Bhd,
74-3, Jalan Wangsa Delima 6, KLSC Wangsa Maju,
53300 Kuala Lumpur, Malaysia.

Date of Revision

December 2022